



Management and Cost Accounting – Budgeting & Variance Analysis

Exercise 5.2

- a) Drawing on exhibit 14.3 (BHDR, p. 430), explain the sequence of steps in preparing a master budget.
- b) Discuss and explain cross-departmental interdependencies in this regard. Which potential challenges might come along with such interdependencies?
- c) Discuss potential tensions or dysfunctional consequences that may arise when firms simultaneously use budgets both for different purposes.
- d) Management accountants are often responsible for coordinating firms' budgeting processes and for consolidating the inputs provided by different managers. Put yourself in the shoes of a management accountant who makes the following observation: Sales managers seem to provide numbers that are considerably lower than what these managers honestly believe is achievable.
 - a) How can such an observation be explained?
 - b) What would you do in such a situation?

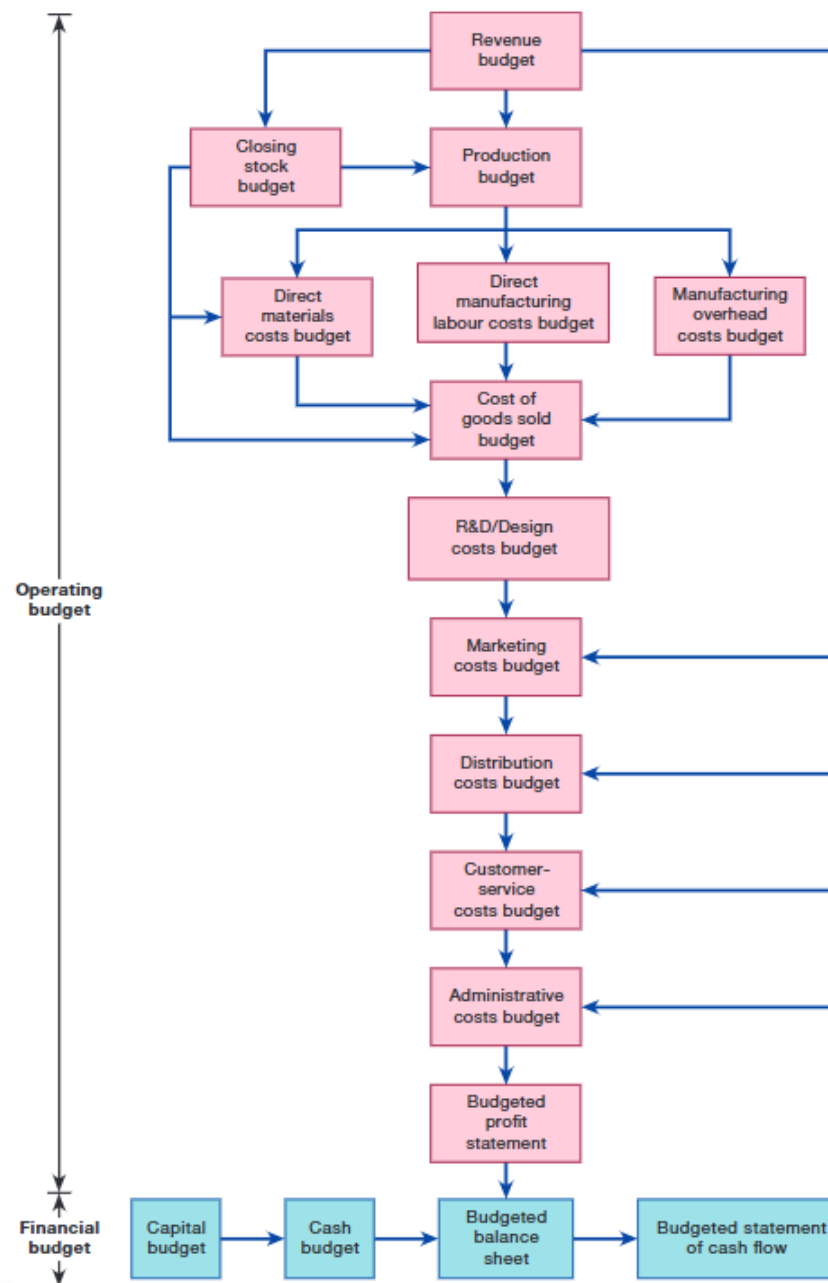
Exercise 5.2

- a) Drawing on exhibit 14.3 (BHDR, p. 430), explain the sequence of steps in preparing a master budget.

Definition of a budget:

“A budget is a quantitative expression of a proposed plan of action by management for a future time period and is an aid to the coordination and implementation of the plan” (BHDR, 2023, p. 423).

The **masterbudget** is a comprehensive financial plan that integrates all of a company's individual budgets into a unified document.

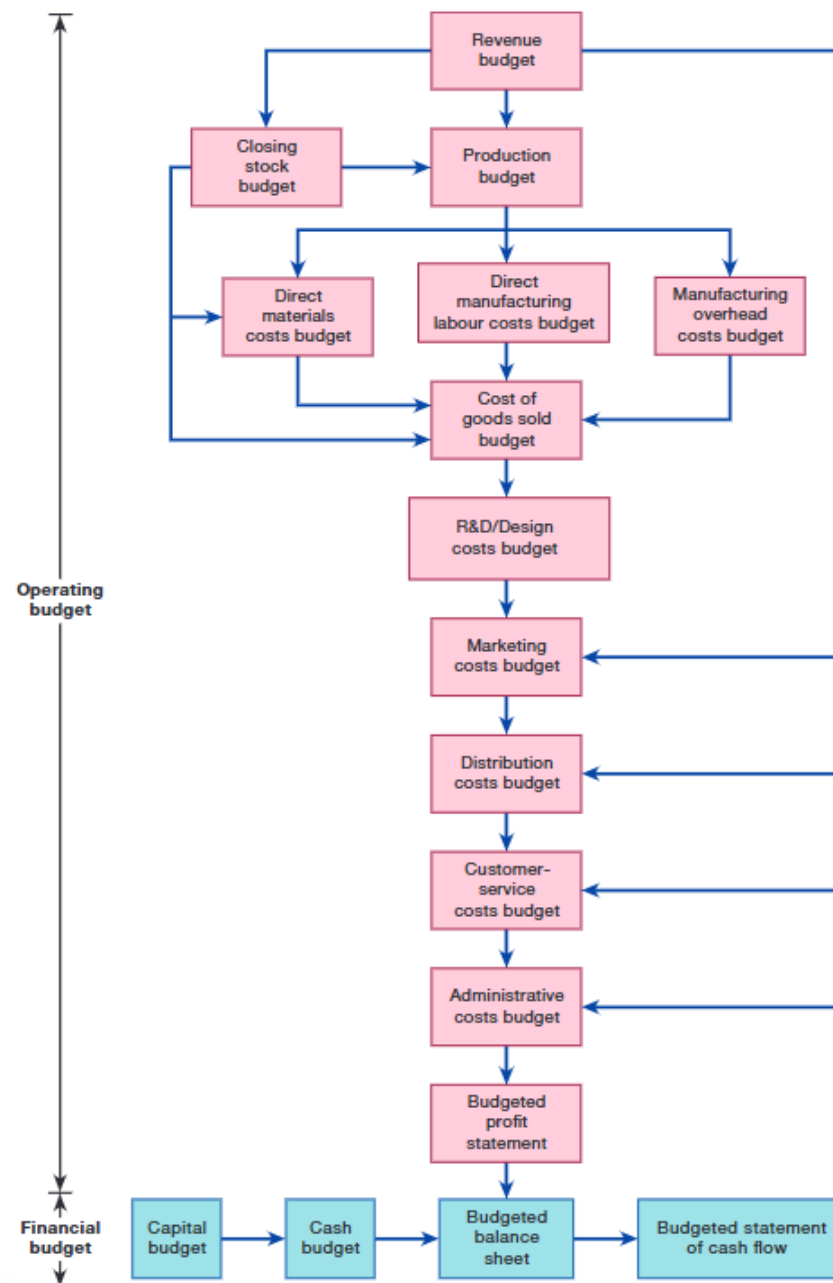


Operating decisions are about the acquisition and use of scarce resources.

Financing decisions deal with how to obtain the funds to acquire resources (BHDR, 2023, p. 423).

Exercise 5.2

- a) Drawing on exhibit 14.3 (BHDR, p. 430), explain the sequence of steps in preparing a master budget.



(1) Sales budget

- Foundation of the master budget
- It forecasts the expected sales volume and revenue
- Factors considered: *market demand, historical sales trends, pricing strategies, etc.*
- Production depends on the forecast level of revenue!

(2) Production budget

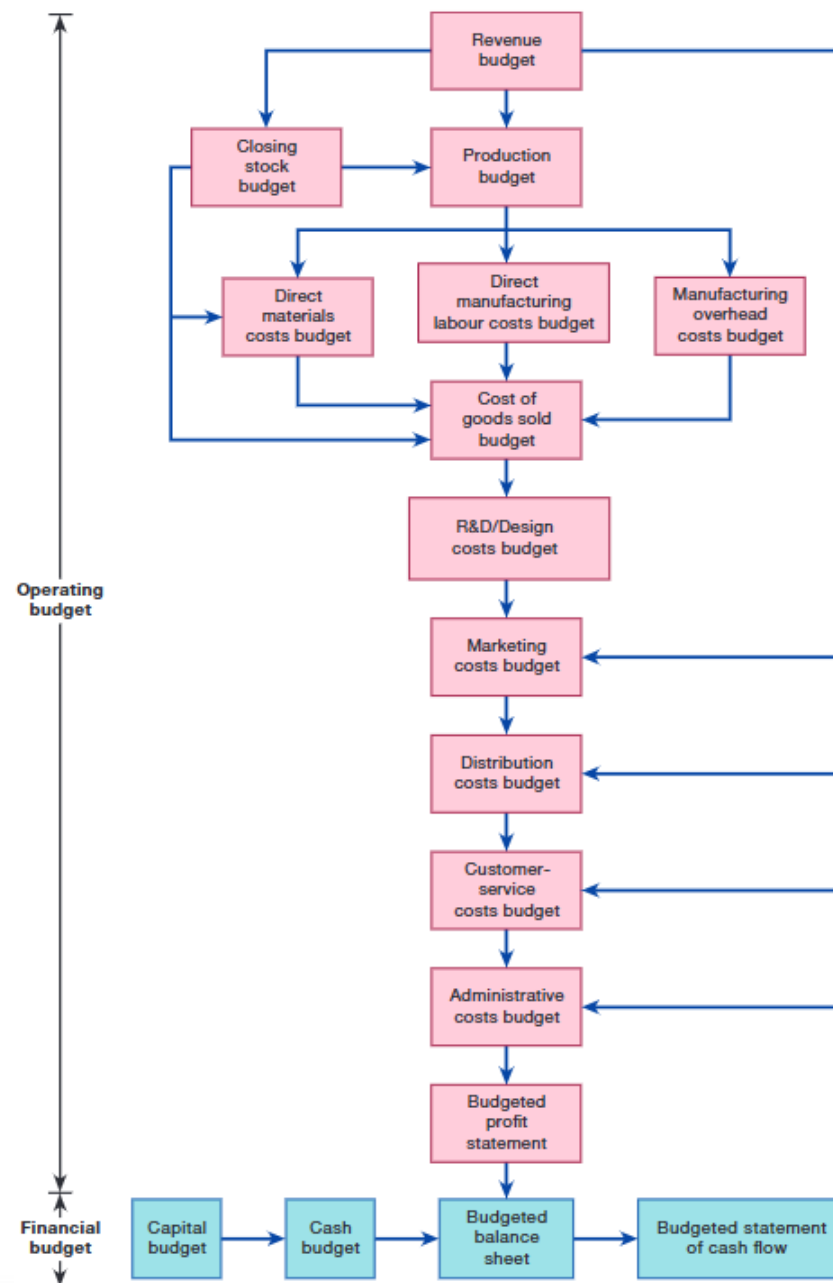
- Based on the sales forecast, this budget determines the number of units that need to be produced

(3) Direct materials & labour budget

- Quantity and costs of raw material required for production
- Estimation of direct labour hours & costs required for production

Exercise 5.2

- a) Drawing on exhibit 14.3 (BHDR, p. 430), explain the sequence of steps in preparing a master budget.



(4) ...

(5) Budgeted operating statements

- Summarises projected revenues and expenses for the budget period
- Based on the revenue budget, COGS budget, and other non-production costs (R&D, marketing, distribution, etc. the budgeted operating statement can be created

(6) Budgeted balance sheet

- Projects the company's financial position at the end of the budget period
- Shows estimated assets, liabilities, and equity.

(7) Budgeted statement of cash flow

- Summarises the expected cash in- and outflows ...

Exercise 5.2

b) Discuss and explain cross-departmental interdependencies in this regard. Which potential challenges might come along with such interdependencies?

- Constructing a master budget is not a siloed exercise confined to a single department (as we have seen before).
- Involves close collaboration and information sharing across various departments.

Potential challenges:

- Information silos, lack of collaboration, conflicting priorities (e.g., the sales department fosters a new product line, which requires a substantial investment vs. the finance department prioritizes reducing debt)
- Inconsistent data sources, data quality issues, etc.
- Time & resource constraints (e.g., tight deadlines)

Exercise 5.2

c) Discuss potential tensions or dysfunctional consequences that may arise when firms simultaneously use budgets both for different purposes.

- **Control *and* motivation**
- Budgets can be used for control (e.g., monitoring performance and enforcing accountability) and motivation (e.g., setting targets and providing incentives)
- Gaming behaviour & the creation of 'budgetary slack'
 - Forecasting higher resources than actually required
 - Forecasting lower profits/revenues than actually expected)
 - Too much slack leads to lower-than-possible performance
- **Short-term *and* long-term**
 - Focusing on short-term financial performance can discourage investments in long-term projects (e.g., investments in R&D)

Exercise 5.2

d) Management accountants are often responsible for coordinating firms' budgeting processes and for consolidating the inputs provided by different managers. Put yourself in the shoes of a management accountant who makes the following observation: Sales managers seem to provide numbers that are considerably lower than what these managers honestly believe is achievable.

(1) How can such an observation be explained?

- Sales managers may deliberately underestimate sales to create a “cushion” for potential shortfalls. This allows them to exceed their targets, which can lead to bonuses, promotions, or other rewards.
- Lower targets may reduce pressure and allow sales manager to focus on other priorities.
- Past budget submissions were met with unrealistic expectations or severe consequences for not meeting targets, and sales managers became hesitant to provide accurate forecasts.
- Etc.

Exercise 5.2

d) Management accountants are often responsible for coordinating firms' budgeting processes and for consolidating the inputs provided by different managers. Put yourself in the shoes of a management accountant who makes the following observation: Sales managers seem to provide numbers that are considerably lower than what these managers honestly believe is achievable.

a) What would you do in such a situation?

- Do not use input from subordinates when setting targets
- Critically check the input from subordinates
- Provide an incentive for accurate forecasts (e.g., **truth-inducing pay schemes**)
 - Reward both: The accuracy of the forecast and the actual performance
- Compare the sales targets provided by the managers with industry benchmarks and the performance of similar companies (if available)

Exercise 15.23

15.23 Price and efficiency variances, problems in standard setting, benchmarking (30–40 minutes)

Poitou-Chemises SARL manufactures shirts for retail chains. Armand Plessis, the accountant, is becoming increasingly disenchanted with Poitou-Chemises' six-month-old standard costing system. The budgeted amounts for both its direct materials and direct manufacturing labour are drawn from its standard costing system. The budgeted and actual amounts for July 2022 were:

	Budgeted	Actual
Shirts manufactured	4 000	4 488
Direct materials cost	€20 000	€20 196
Direct materials units used (rolls of cloth)	400	408
Direct manufacturing labour costs	€18 000	€18 462
Direct manufacturing labour-hours	1 000	1 020

There were no opening or closing stocks of materials.

Armand observes that in the past six months he has rarely seen an unfavourable variance of any magnitude. The standard costing system is based on a study of the operations conducted by an independent consultant. He decides to play detective and makes some unobtrusive observations of the workforce at the plant. He notes that even at their current output levels, the workers seem to have a lot of time to discuss football, sitcoms and cooking recipes.

At a recent industry conference on 'Benchmarking and Competitiveness', Françoise Daubigné, the accountant of Textiles-Georges-Grassens, told Armand that Textiles-Georges-Grassens had employed the same independent consultant to design a standard costing system. However, the company dismissed him after two weeks because Textiles-Georges-Grassens employees quickly became aware of the consultant observing their work.

At the industry conference, Armand participated in seminars on 'benchmarking for the fabric industry'. A consultant for France-Solutions SA showed how she could develop six-month benchmark reports on the estimated costs of Poitou-Chemises' major competitors. She indicated that she was already examining the estimated cost of shirts manufactured by the four largest importers into Europe. These importers had taken much business from Poitou-Chemises in recent years. This information would soon be available by subscribing to the France-Solutions monthly service.

Required

- 1 Calculate the price and efficiency variances of Poitou-Chemises for direct materials and direct manufacturing labour in July 2022.
- 2 Describe the types of action the employees at Textiles-Georges-Grassens may have taken to reduce the accuracy of the standards set by the independent consultant. Why would employees take those actions?
Is this behaviour ethical?
- 3 Describe how Poitou-Chemises might use information from France-Solutions when computing the variances in requirement 1.
- 4 Discuss the pros and cons of Poitou-Chemises using the France-Solutions information to increase its cost competitiveness.

(BHDR, 2023, pp. 487-488)

Solution exercise 15.23

Task 1: Calculate the price and efficiency variances of Poitou-Chemises for direct materials and direct manufacturing labour in July 2022.

	Budgeted	Actual	Flexible Budget 1	Flexible Budget 2
Shirts manufactured	4,000	4,488		
Direct materials cost	€ 20,000	€ 20,196		
Direct materials units used (rolls of cloth)	400	408		
Direct manufacturing labour costs	€ 18,000	€ 18,462		
Direct manufacturing labour hours	1,000	1,020		
Total cost	€ 38,000	€ 38,658		

SBV = 658 € (u)

“Flexing” the budget allows us to learn more about the origins of an overall variance. On the cost side, this is called the Static Budget Variance (SBV).

Each column (Budgeted, Actual, FB1, FB2) represents a different combination of **(1)** output volume (here: shirts), **(2)** prices (here: for material and labour), **(3)** efficiency (here, material per shirt, hours of labour work per shirt).

FB1: Actual output volume, budgeted prices, budgeted efficiency rates.

FB2: Actual output volume, budgeted prices, actual efficiency rates.

Efficiency variance: FB1 – FB2

Price variance: FB2 – actual cost

Solution exercise 15.23

Task 1: Calculate the price and efficiency variances of Poitou-Chemises for direct materials and direct manufacturing labour in July 2022.

	Budgeted	Actual
Shirts manufactured	4,000	4,488
Direct materials cost	€ 20,000	€ 20,196
Direct materials units used (rolls of cloth)	400	408
Direct manufacturing labour costs	€18,000	€18,462
Direct manufacturing labour hours	1,000	1,020

Step 1

Budgeted materials input/shirt	$4,000/400 = 0,1$ roll of cloth/shirt
Budgeted manufacturing labour hours/shirt	$1,000/4.000 = 0,25$ hours/shirt
Budgeted materials cost/roll	$€20,000/400 = €50$ per roll
Budgeted manufacturing labour/hour	$€18,000/1,000 = €18$ per hour
Actual shirt manufactured	4,488 shirts

Step 2

	Budgeted	Actual	Flexible Budget 1	Flexible Budget 2
Shirts manufactured	4,000	4,488	4,488	4,488
Direct materials cost	€ 20,000	€ 20,196	$0,1 * € 50 * 4.488 = € 22,440$	$408 * € 50 = € 20,400$
Direct materials units used (rolls of cloth)	400	408	$4488 * 0,1 = 448.8$	408
Direct manufacturing labour costs	€ 18,000	€ 18,462	$0,25 * € 18 * 4.488 = € 20,196$	$1.020 * € 18 = € 18,360$
Direct manufacturing labour hours	1,000	1,020	$4488 * 0,25 = 1,122$	1,020
Total cost	€ 38,000	€ 38,658	€ 42,636	€ 38,760

Solution exercise 15.23

Task 1: Calculate the price and efficiency variances of Poitou-Chemises for direct materials and direct manufacturing labour in July 2022.

	Budgeted	Actual	Flexible Budget 1	Flexible Budget 2
Shirts manufactured	4,000	4,488	4,488	4,488
Direct materials cost	€ 20,000	€ 20,196	$0,1 * € 50 * 4.488 = € 22.440,00$	$408 * € 50 = € 20.400,00$
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Direct manufacturing labour hours	1,000	1,020	$4488 * 0,25 = 1,122$	1020
Total cost	€ 38,000	€ 38,658	€ 42,636	€ 38,760
Output:	Budget	Actual	Actual	Actual
Price:	Budget	Actual	Budget	Budget
Efficiency:	Budget	Actual	Budget	Actual

Efficiency Variance (FB1 – FB2)

Total: 3 876 € (f)

Thereof material: 2 040 € (f)

Thereof labour: 1 836 € (f)

Price Variance (FB2 – Actual cost)

Total: 102 € (f)

Thereof material: 204 (f)

Thereof labour: 102 (u)

Solution exercise 15.23

Task 1: Calculate the price and efficiency variances of Poitou-Chemises for direct materials and direct manufacturing labour in July 2022.

	Actual costs incurred (Actual input × Actual price)	Actual input × Budgeted price	Flexible budget (Budgeted input allowed for actual Output achieved × Budgeted price)
Direct materials	€20,196	$(408 \times € 50)$ €20,400	$(4,488 \times 0.10 \times €50)$ €22,440
	↑	↑	↑
	€204 F	€2,040 F	
	Price variance	Efficiency variance	
Direct manufacturing labour	€18,462	$(1,020 \times €18)$ €18,360	$(4,488 \times 0.25 \times €18)$ €20,196
	↑	↑	↑
	€102 U	€1,836 F	
	Price variance	Efficiency variance	

F ... favourable effect on operating profit

U ... unfavourable effect on operating profit

→ The flexible budget variance of € 2.244 for direct materials is attributed to € 204 price variance and € 2.040 to efficiency variance

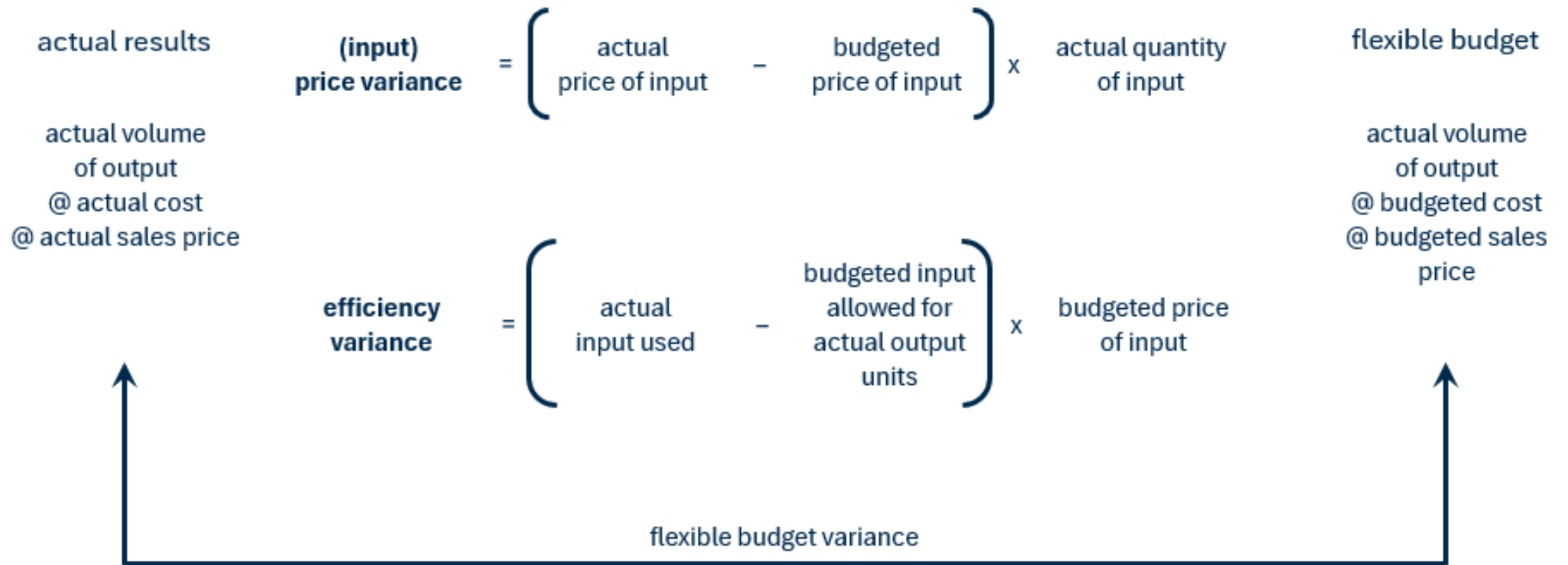
→ Put simply:

→ Costs were lower than expected

→ € 204 less costs due to price changes

→ € 2.040 less costs due to efficiency improvements

Different approach



Solution exercise 15.23

Task 2: Describe the types of action the employees at Textiles-Georges-Grassens may have taken to reduce the accuracy of the standards set by the independent consultant. Why would employees take those actions? Is this behaviour ethical?

Actions employees may have taken:

- a) Adding steps in working on a shirt that are not necessary.
- b) Taking more time on each step than is necessary.
- c) Creating problem situations so that the budgeted amount of average downtime will be overstated.
- d) Etc.

Employees may take these actions for several possible reasons:

- a) They may be paid on a piece-rate basis with incentives for above-budgeted production.
- b) They may want to create a relaxed work atmosphere, and less demanding standards can reduce stress
- c) They may have 'them vs. us' mentality rather than a partnership perspective.
- d) Etc.

Solution exercise 15.23

Task 3: Describe how Poitou-Chemises might use information from France-Solutions when computing the variances in requirement 1.

- a) For pricing and product emphasis purposes, Poitou-Chemises should avoid getting into a pricing war with a competitor which has a sizeably lower cost structure.
- b) As indicators of areas where Poitou-Chemises is either highly cost competitive or highly cost non-competitive.
- c) As performance targets for motivating and evaluating managers.
- d) Etc.

Solution exercise 15.23

Task 4: Discuss the pros and cons of Poitou-Chemises using the France-Solutions information to increase its cost competitiveness.

Main pros:

- Highlights to Poitou-Chemises in a direct way how it may or may not be cost competitive.
- Provides a 'reality check' to many internal positions about efficiency or effectiveness.

Main cons:

- Poitou-Chemises may not be comparable to companies in the database.
- Data about other company's costs may not be reliable.
- Cost of France-solutions reports.

Questions?



<https://www.pinterest.de/pin/799037158920259827/>

References

Bhimani, A., Horngren, C. T., Datar, S. M., & Rajan, M. V. (2023). *Management and Cost Accounting* (8th ed.). Pearson.